

EAGLE Compliance Matrix Flow

Where the matrix lives today, and where the 4/16 meeting fixes would plug in

This deck maps how `eagle-plugin/data/matrix.json` feeds every tool in the EAGLE stack — and shows where the two proposals from the April 16 EAGLE-vs-Research-Optimizer review would attach to that existing flow. The goal: keep rules in data (queryable, enforceable) rather than in prompt text (model-interpreted).

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SCENARIO 1 OF 5

Current State

`matrix.json` is loaded once at import time in `server/app/compliance_matrix.py`. Four tools read from it today:
`query_compliance_matrix`, `manage_package`, `research`, and `search_far`.

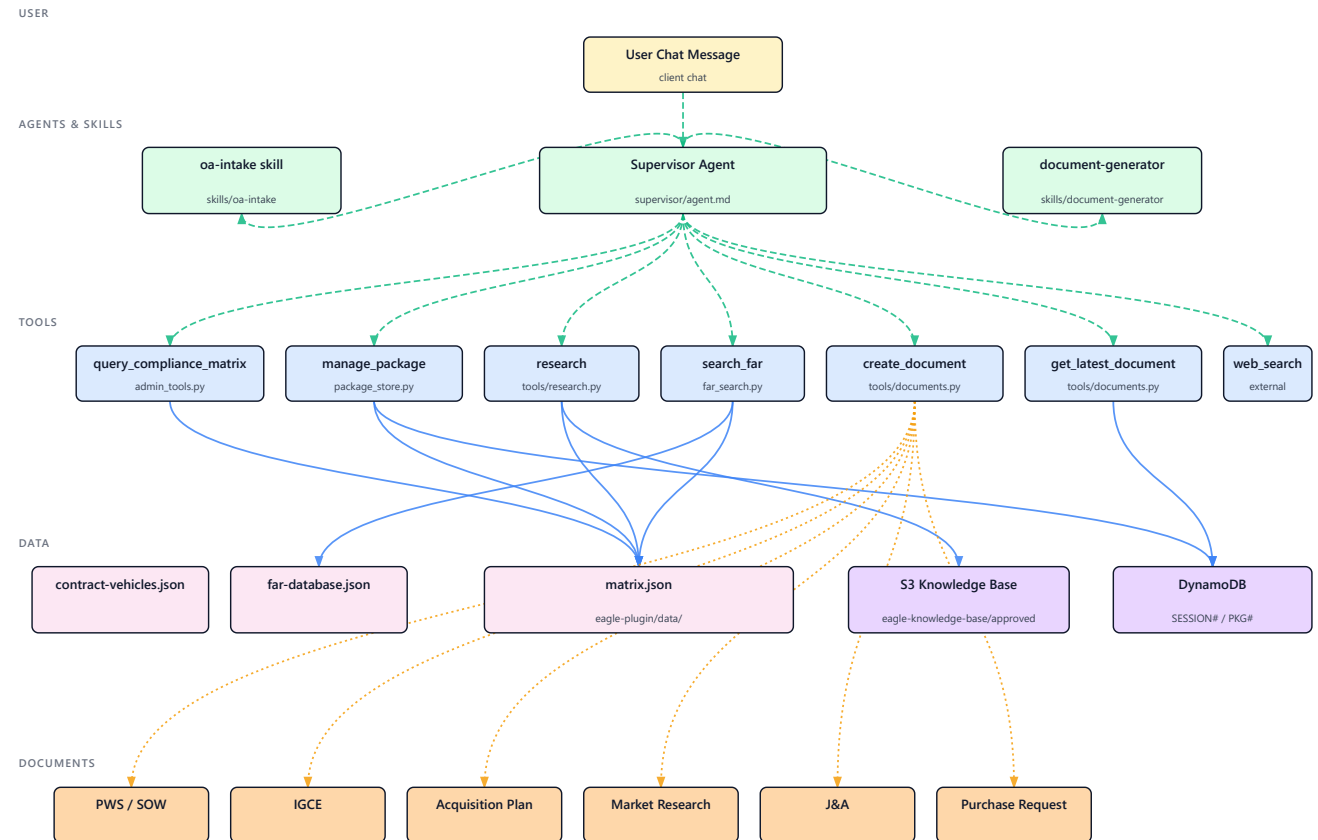
What's already in `matrix.json`

`thresholds` · `contract_types` · `doc_rules` ·
`approval_chains` · `warnings` · `hhs_nih_docs`

The hole: `create_document` has no read-arrow into the matrix. It accepts whatever the supervisor sends, with no validation that intake facts were actually collected. That's where docs get generated too early.

Rules that live in prompt (not data):

"DEFAULT TO ACTION", "CHECKLIST-FIRST", "DO THE WORK". These are model-interpreted — the meeting's "should never ask" behaviors are exactly what prompt-only rules fail to prevent.



--- tool call
— reads from matrix
... emits document

SCENARIO 2 OF 5

Meeting Problems

Three concrete failures from the 4/16 review, mapped to the places where they originate in the flow.

1. PWS before cadence

EAGLE generated a PWS before confirming training-event frequency. Intake was incomplete and no gate enforced completeness.

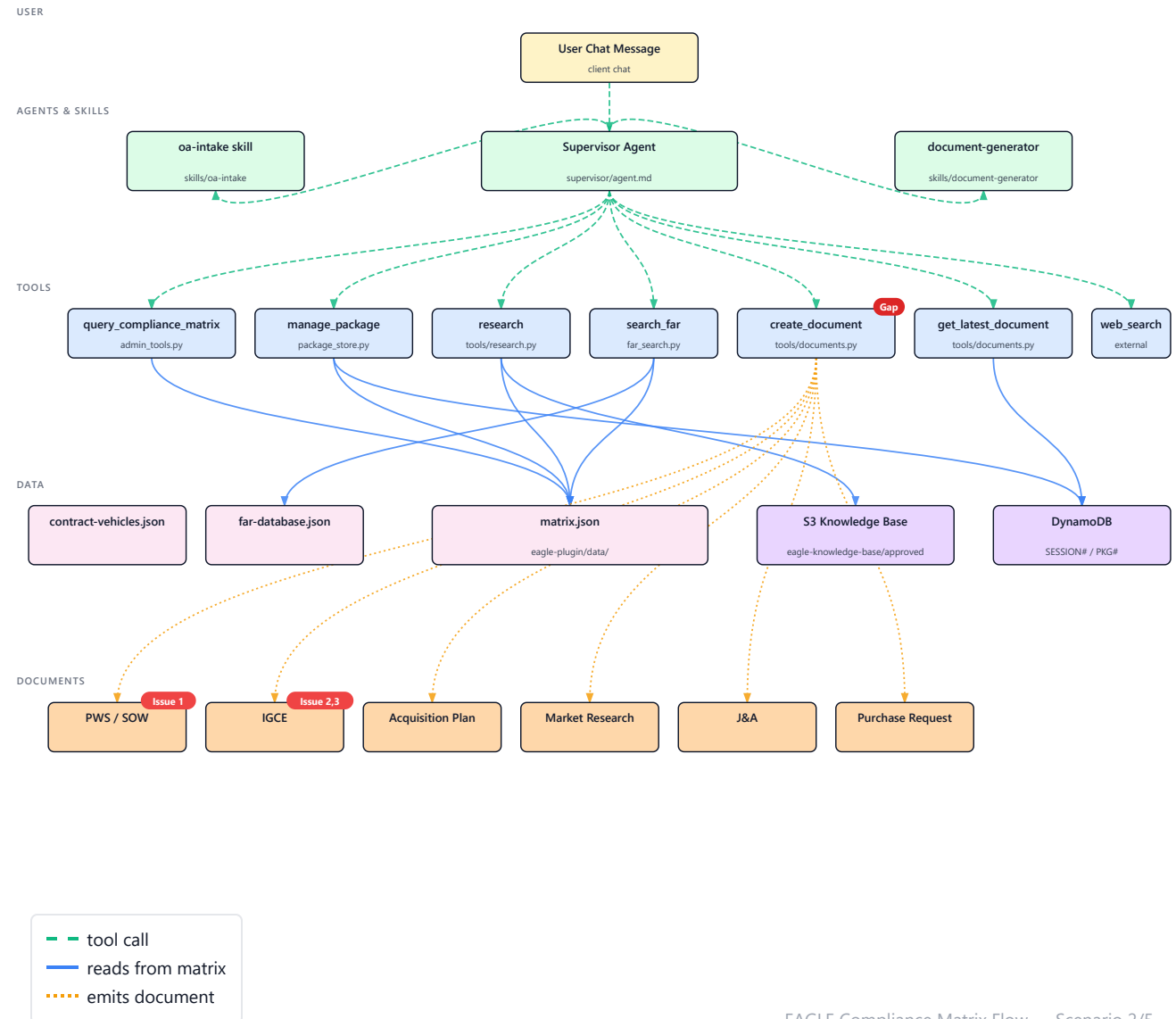
2. Budget-vs-IGCE reconcile

"Your budget is \$2M but IGCE is \$450K, what do you want?" Ryan: "That question should never be asked."

3. Ceiling ballooning

Research Optimizer inflated to \$2M when given a budget range. The model treated budget as a target instead of a not-to-exceed ceiling.

All three originate between the supervisor's decision to call `create_document` and the actual document being emitted — the gap where no data-backed rule exists today.



SCENARIO 3 OF 5

Proposal 1 — Intake Gate

Add a new matrix key
`intake_required_facts` — a per-doc-type list of intake fields that must be present before generation. Add a PRE-GENERATION INTAKE GATE section to `supervisor/agent.md` that reads it and blocks `create_document` until facts are collected.

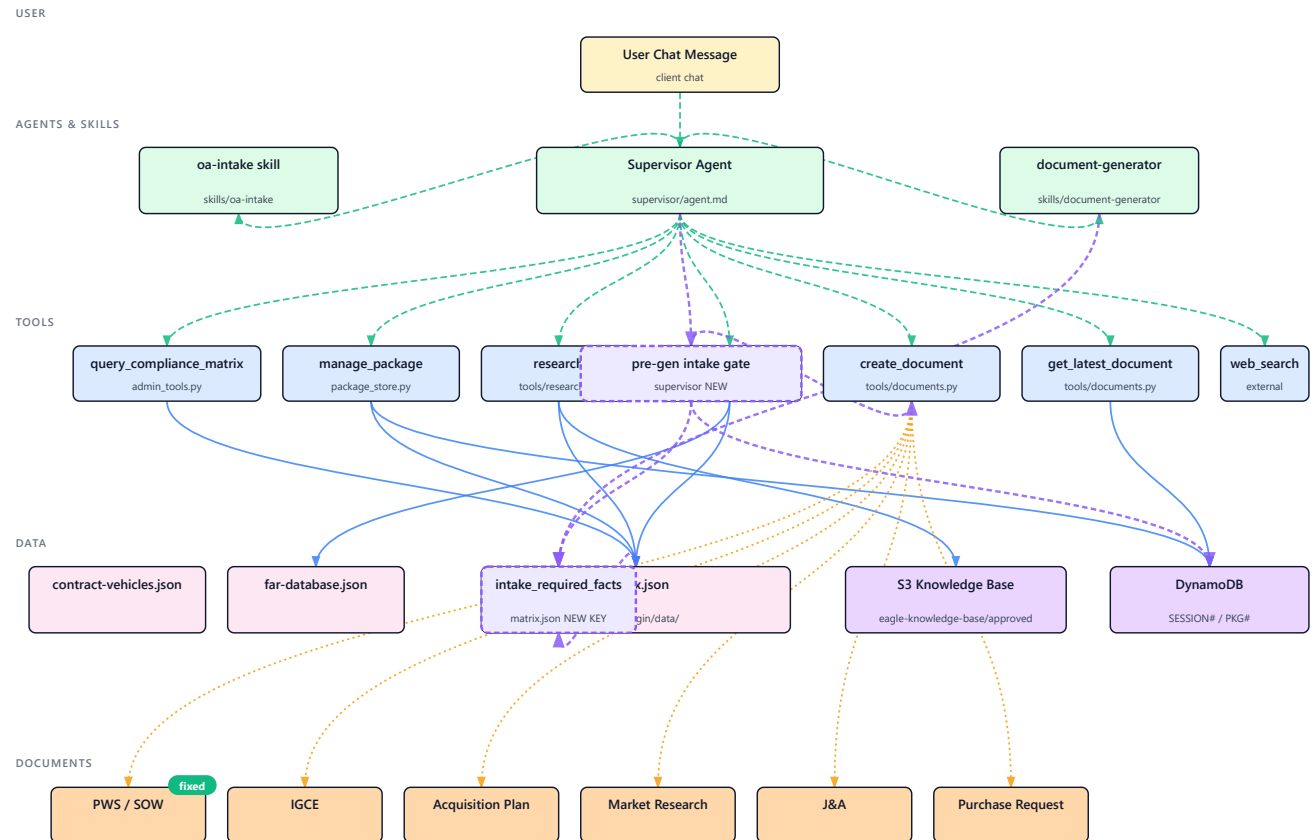
`matrix.intake_required_facts (new)`

pws: [scope, pop, place, event_cadence, deliverable_format]
 sow: [scope, pop, place, task_breakdown]
 igce: [scope, pop, labor_cats, budget_ceiling]
 mrr: [scope, naics, estimated_value_range]
 acquisition_plan: [scope, igce_value, contract_type, competition]
 ja: [scope, proposed_contractor, far_6_302_x, sole_source_rationale]

The gate batches missing facts into ONE clarifying question, not a drip. Matches `oa-intake`'s "ask 2–3 at a time" philosophy.

Fixes issue 1

`intake_required_facts.pws.event_cadence` blocks PWS generation until cadence is confirmed.



- tool call
- reads from matrix
- ... emits document
- - proposed (new)

Proposal 2 — Budget Semantics

Add a matrix key `budget_semantics` that codifies Ryan's rules as data. Add a BUDGET SEMANTICS RULE section to `supervisor/agent.md` that reads it before any cost-related output.

`matrix.budget_semantics (new)`

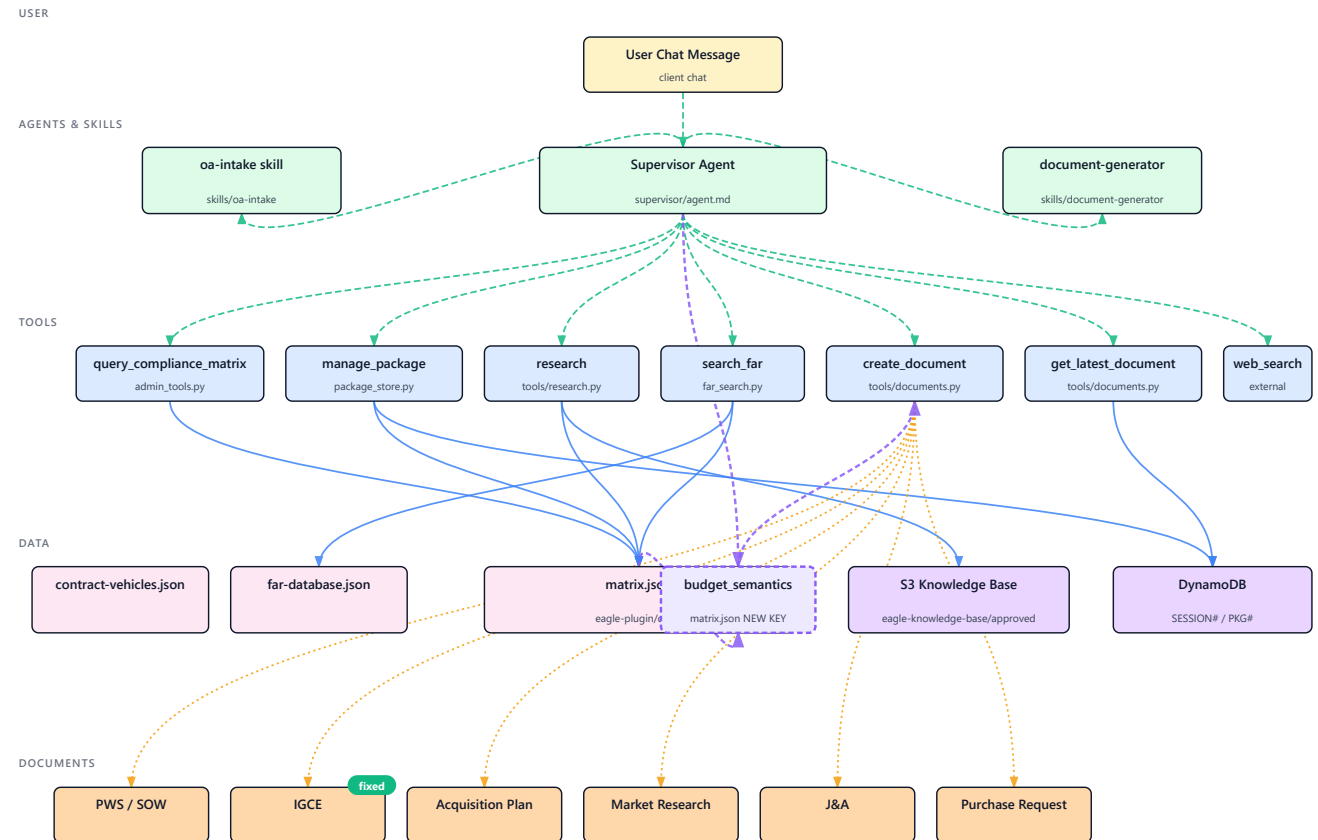
```
budget_is_ceiling: true
igce_is_estimated_value: true
propagation: "IGCE auto-propagates as
estimated_value to AP, J&A, SSP"
forbidden_behaviors: [
  "asking user to reconcile IGCE vs budget",
  "suggesting scope expansion to consume
remaining budget",
  "inflating quantities/rates to reach a budget
target"
]
```

Fixes issue 2

`forbidden_behaviors[0]` directly forbids the reconcile question.

Fixes issue 3

`budget_is_ceiling` + `forbidden_behaviors[2]` stop ceiling ballooning.



SCENARIO 5 OF 5

Target State

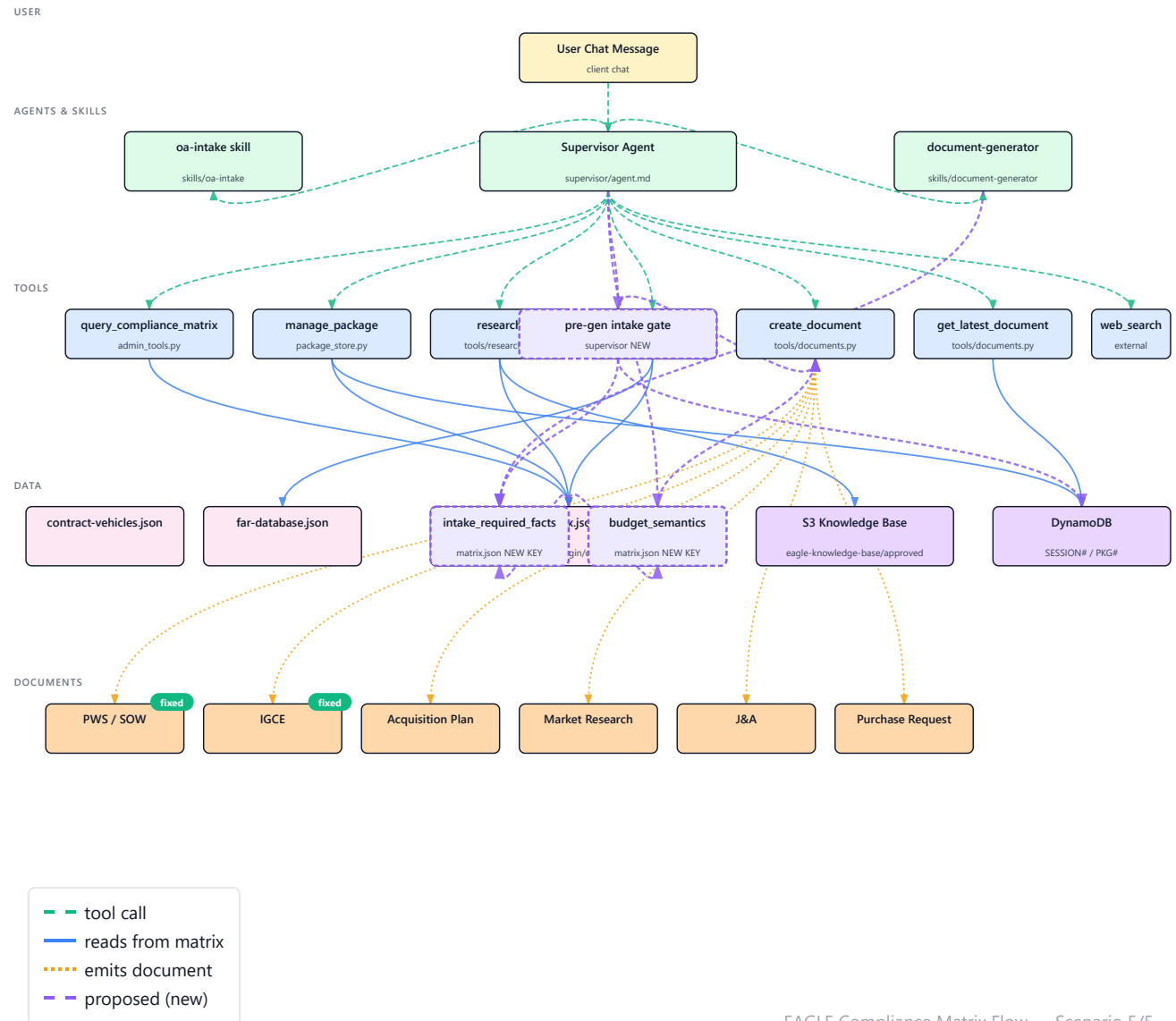
Both proposals wired in. Matrix is still the single source of truth — all new rules live as data, all tools can read them, the supervisor prompt just points at matrix keys.

Changes required

matrix.json: +2 top-level keys
 supervisor/agent.md: +2 sections (~25 lines each)
 document-generator/SKILL.md: +1 table column
 oa-intake/SKILL.md: trim hardcoded Phase 2 lists
 backend: +1 validator in create_document dispatch

Why matrix, not prompt? The backend validator can read matrix.json; it can't read prompt text. That means the rule is enforced in code (blocks the bad call), not just suggested in prompt (which the model can override under pressure).

Rollback cost: revert 1 JSON file + 2 prompt sections. Small blast radius.



Fix Matrix

Meeting issue → matrix.json key → supervisor section

Meeting Issue (4/16)	Today's Cause	New matrix.json Key	Supervisor Section
PWS generated before training-event cadence confirmed	No intake-completeness check before <code>create_document</code>	<code>intake_required_facts.pws.event_cadence</code>	PRE-GENERATION INTAKE GATE — reads matrix, batches missing facts into one question, blocks generation
"Your budget is \$2M but IGCE is \$450K, what do you want?"	Prompt asks user to reconcile; no rule forbids it	<code>budget_semantics.forbidden_behaviors[0]</code>	BUDGET SEMANTICS RULE — explicitly forbids the reconcile question
RO ballooned IGCE to \$2M ceiling when given a budget range	Budget treated as target, not ceiling	<code>budget_semantics.budget_is_ceiling</code> + <code>forbidden_behaviors[1,2]</code>	BUDGET SEMANTICS RULE — IGCE is authoritative estimated value; budget is checking-account ceiling only
Stale FAR vs RFO in KB (separate KB sync issue)	S3 knowledge base desync	— (out of scope for matrix)	Action Item #1 — Alvi/WADA diff + re-ingest
Contract type defaulted to "Labor Hour"	Prompt reference; matrix has LH in contract_types	<code>contract_types.lh.discouraged: true</code> (tiny edit)	Existing CONTRACT-TYPE RECOMMENDATION section updated to skip discouraged types
IGCE missing methodology narrative	Template doesn't require it	— (template change, not matrix)	document-generator/SKILL.md — IGCE template section 2 expanded

Summary: Three of six meeting issues are fixed by adding two top-level keys to matrix.json and two sections to supervisor/agent.md. The remaining three are scoped to KB sync (Action #1), contract-type data edit, and document-generator template — none of which expand the matrix proposal's blast radius.